P

Physical Chemistry Cume

Dec 7th 2018

1) The full wavefunction of an atomic orbital can be separated into the angular and the radial parts

$$\psi(\theta, \phi, R) = Y(\theta, \phi)R(r)$$

The radial distribution function is given by $r^2[R(r)]^2$. Sketch the radial distribution functions as a function of r for the hydrogen 3s, 3p and 3d orbitals. Plot them all on the **same** graph, clearly labeling which is which, making sure to pay close attention to their relative peak positions and their relative intensities. You do not have to provide numbers on the axes; just assume that the axes ranges are set to show all the relevant features. **(30 points)**

2) The wavefunction of a 1s orbital for a hydrogenic (hydrogen-like) atom is

$$\psi = Ne^{-Zr/a_0}$$

Where a_o = 52.9 pm

a) Find the normalization constant, N. **(20 points)**

b) For H, He^+ and Li^{2+}, calculate the **average (expectation) distance** of the electron in the 1s orbital from the nucleus, i.e. <r>, using the correct normalized wavefunction. **(20 points)**

c) For H, He^+ and Li^{2+}, calculate the **most probable distance** of the electron in the 1 s orbital from the nucleus. **(20 points)**

d) For H, He^+ and Li^{2+} describe and explain the trends found in parts b) and c) **(10 points)**

Useful information:

Integration over a sphere is given by $\int 4\pi r^2 dr$

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{(n+1)}}$$